MERCon 2026
MORATUWA ENGINEERING RESEARCH CONFERENCE

TARVI: Transcription-Factor Aided RNA Velocity Inference

with Supervised Latent Time and Velocity-Pseudotime Blending

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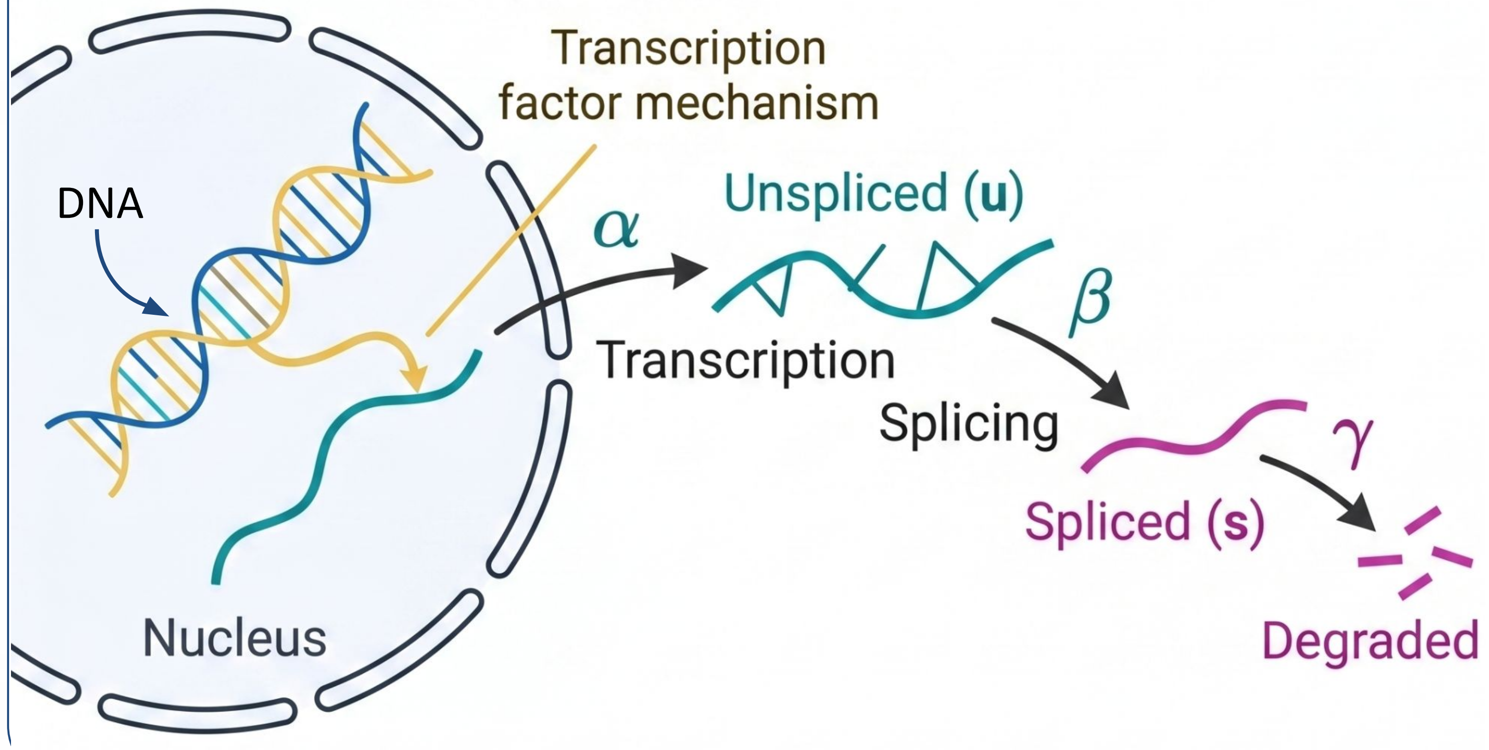
What is **RNA Velocity?**

A technique that **predicts how a cell's gene expression will change in the near future**

Figure out the "**direction**" and "**speed**" of these, **changes in gene counts** for **each individual cell** → showing **where a cell is headed in its biological development**.

Central Dogma of Molecular Biology



Transcription: $$\frac{du}{dt} = \alpha_g(t) - \beta_g u_g$$

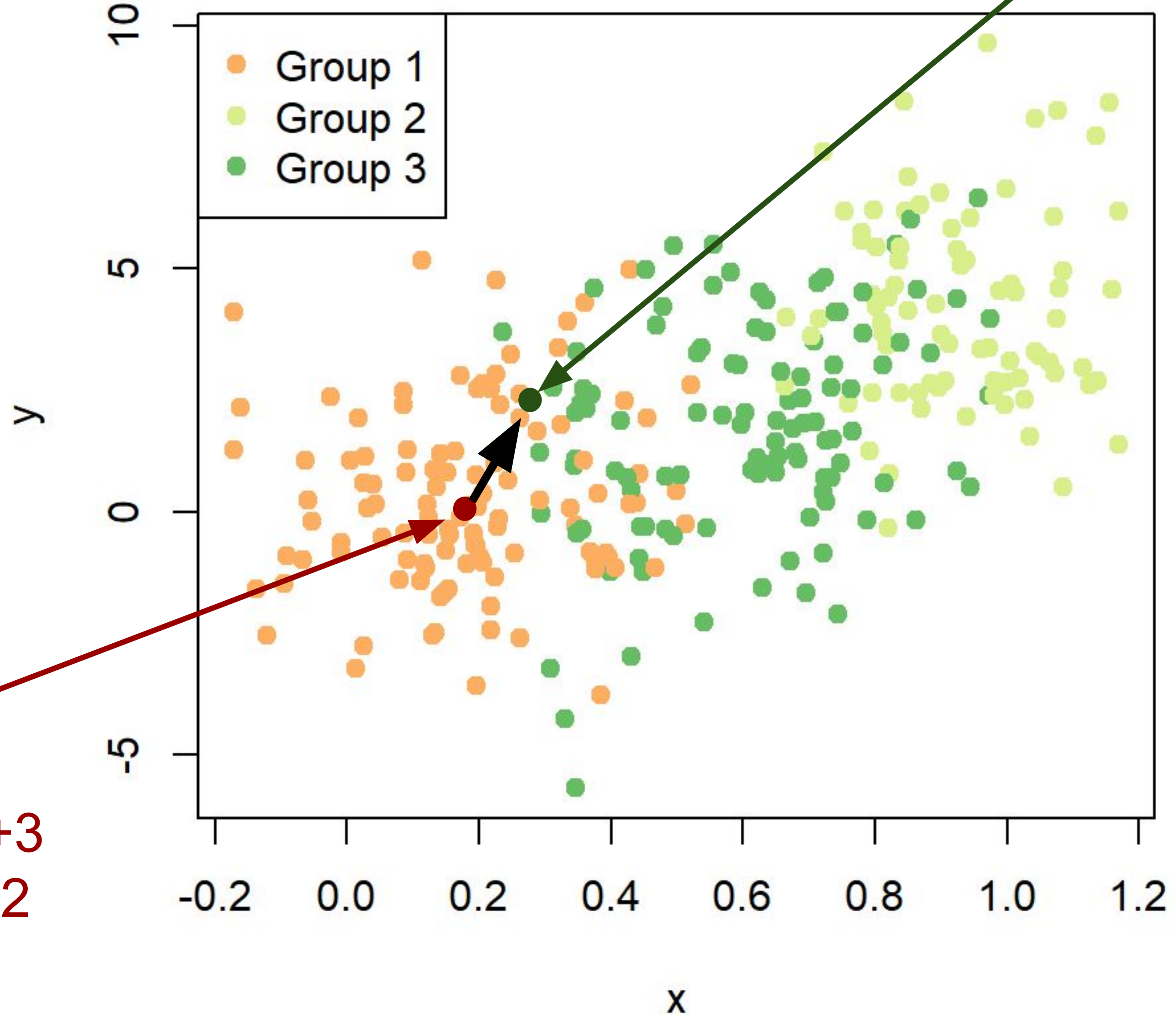
Splicing: $$\frac{ds}{dt} = \beta_g u_g - \gamma_g s_g$$

Velocity

Velocity = Rate of change in spliced RNA count

Simple Example to Understand

Cell_2
Ngene_1 = 23
Ngene_2 = 28



Future_Cell_1
Ngene_1 = 23
Ngene_2 = 28

Cell_1
Ngene_1 = 20, Vgene_1 = +3
Ngene_2 = 30, Vgene_2 = -2

Limitations of current methods

Treated as a static **gene-level constant.**

Ignores that genes are switched on by **transcription factors**, which differ per cell.

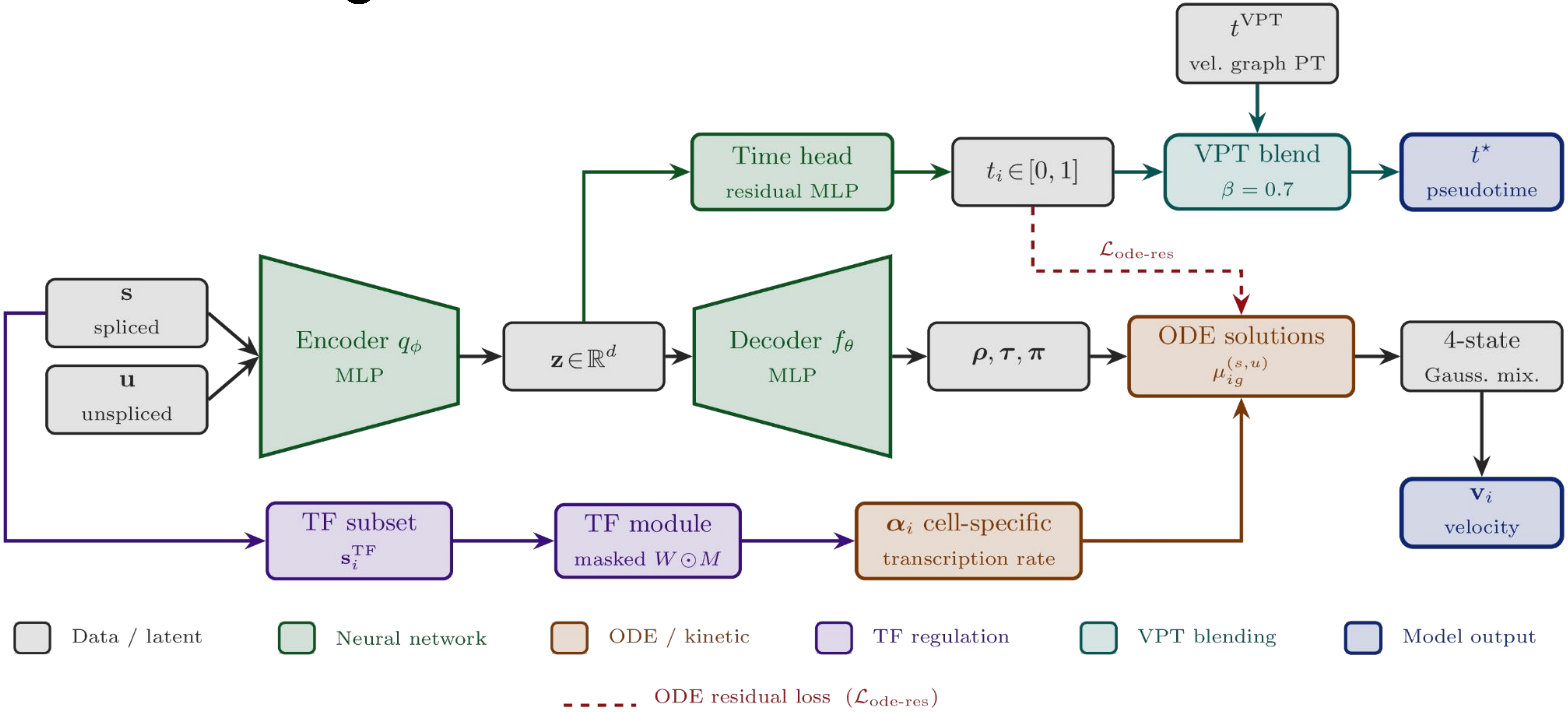
Latent cell time is extracted merely as an uncalibrated by-product of mixture weights

Failing to align with a **true development ordering**

Local velocity and global trajectory estimated **independently**

Produces r**eversed and discontinuous flows**

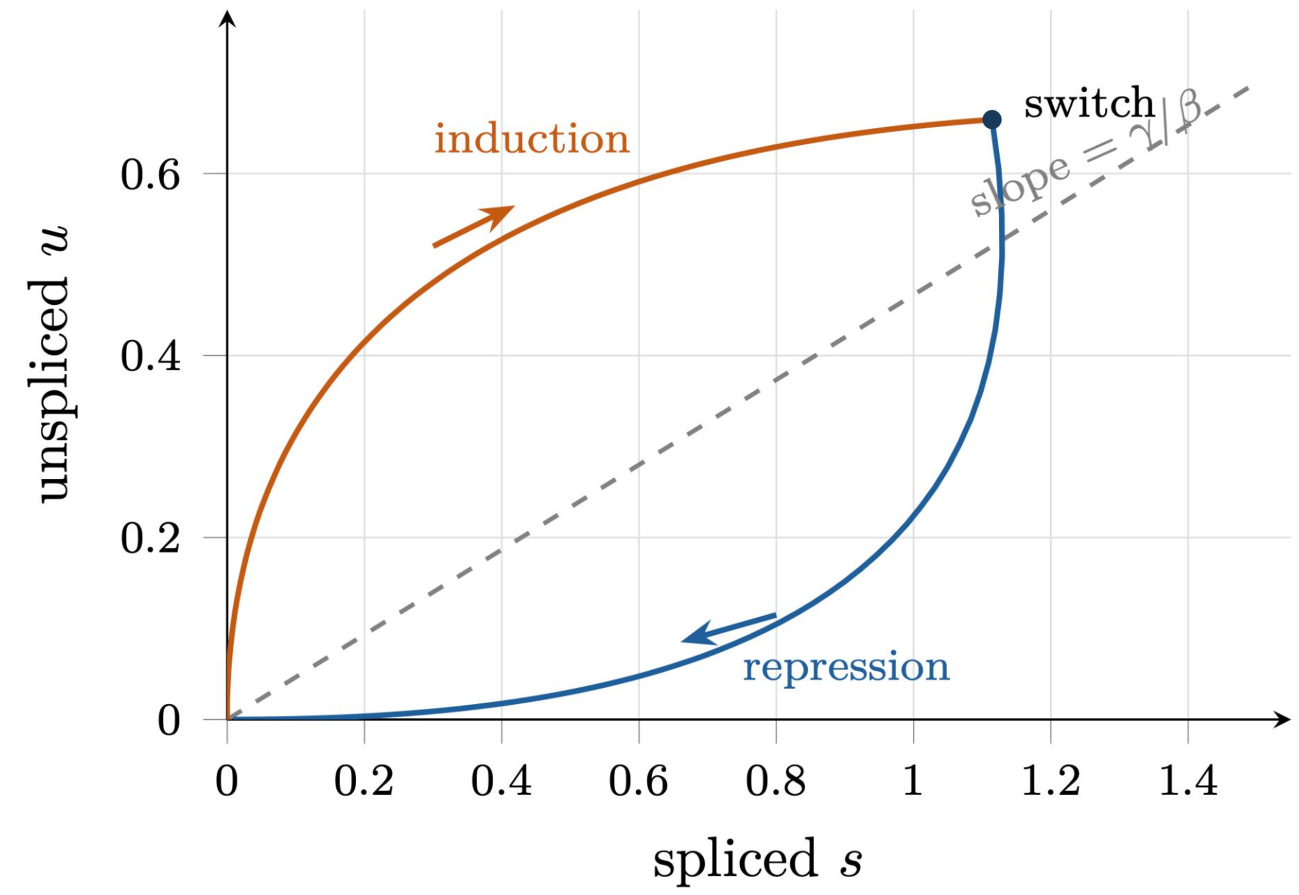
TARVI at a glance



Decoder → four kinetic states

The decoder maps z to three quantities, **per cell and per gene:**

Output	Shape	Meaning
π	$[N, G, 4]$	Dirichlet concentrations over kinetic states (sampled with `rsample`)
ρ	$[N, G] \in (0, 1)$	fractional progress through **induction**
τ	$[N, G] \in (0, 1)$	fractional progress through **repression**



1. Branch Evolution

Induction Branch: starts from $u = s = 0$, evolves over $t_{\text{ind}} = t_s \rho_{i,g}$.

Repression Branch: starts from induction endpoint (u_0, s_0), evolves over $t_{\text{rep}} = (t_{\max} - t_s)\tau_{i,g}$.

2. Closed-Form ODE Solutions

Induction Equations:

$$u_{\text{ind}}(t) = \frac{\alpha}{\beta}\left(1 - e^{-\beta t}\right)$$

$$s_{\text{ind}}(t) = \frac{\alpha}{\gamma}\left(1 - e^{-\gamma t}\right) + \frac{\alpha}{\gamma - \beta}\left(e^{-\gamma t} - e^{-\beta t}\right)$$

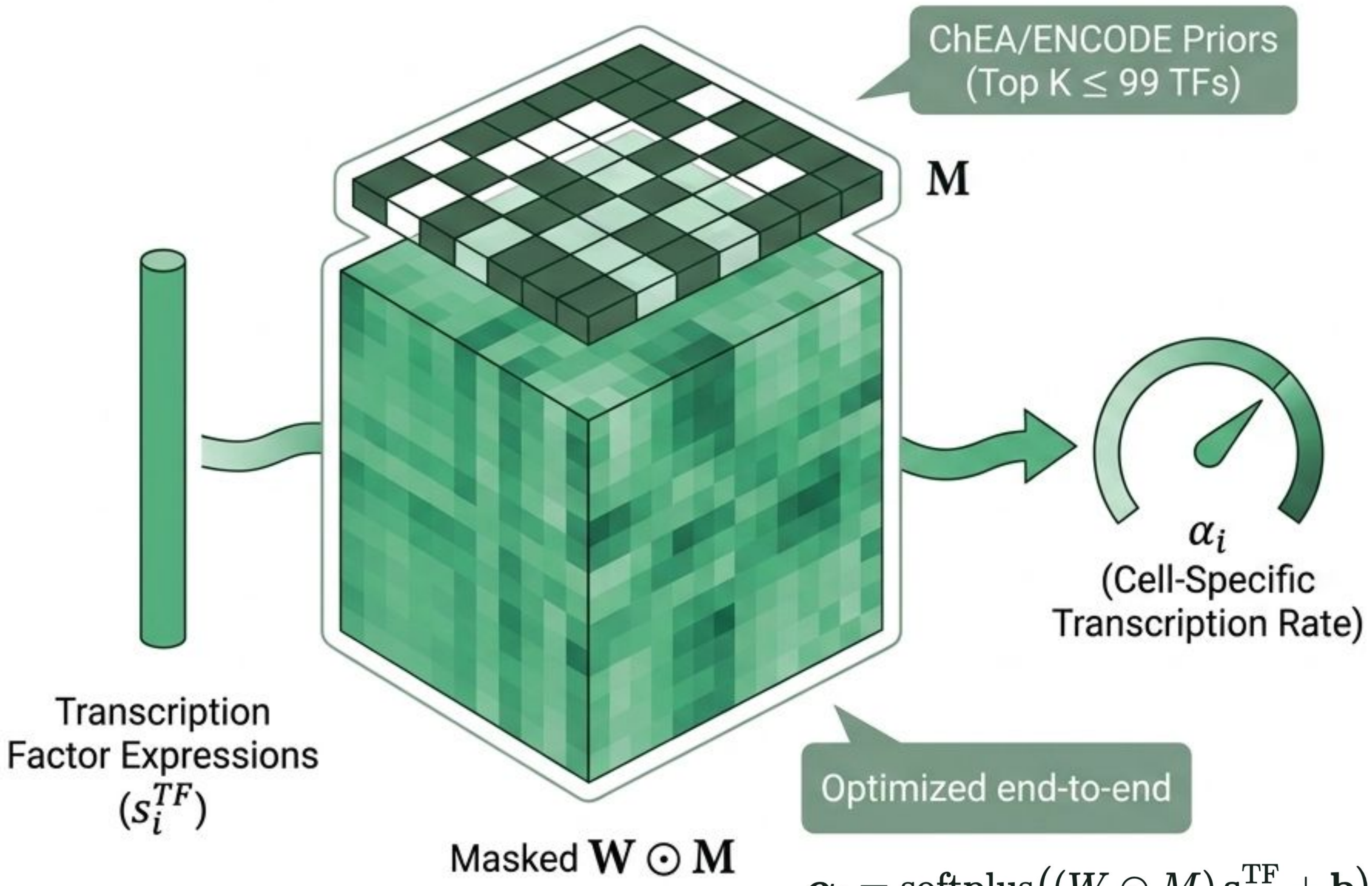
Repression Equations:

$$u_{\text{rep}}(t) = u_0 e^{-\beta t}$$

$$s_{\text{rep}}(t) = s_0 e^{-\gamma t} - \frac{\beta u_0}{\gamma - \beta}\left(e^{-\gamma t} - e^{-\beta t}\right)$$

Contribution 01

Cell-specific, TF-regulated transcription

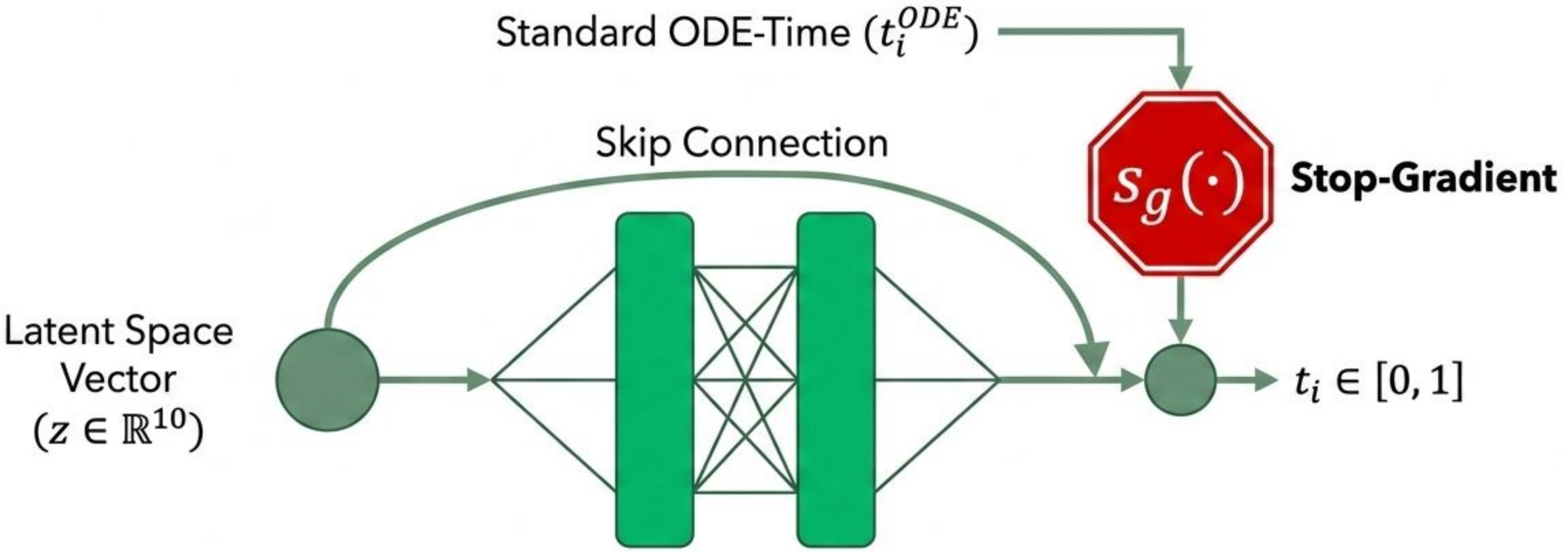


$$\boldsymbol{\alpha}_i = \mathrm{softplus}\big((W \odot M)\,\mathbf{s}_i^{\mathrm{TF}} + \mathbf{b}\big)$$

- Replaces the static gene-level α_g with a unique, cell-specific rate α_i.
- Explicitly driven by biological ground truth via hierarchical Gene Regulatory Networks.
- Unlike TFvelo, this module is integrated directly into a generative model and jointly optimized against the ODE likelihood.

Contribution 02

Supervised residual time head

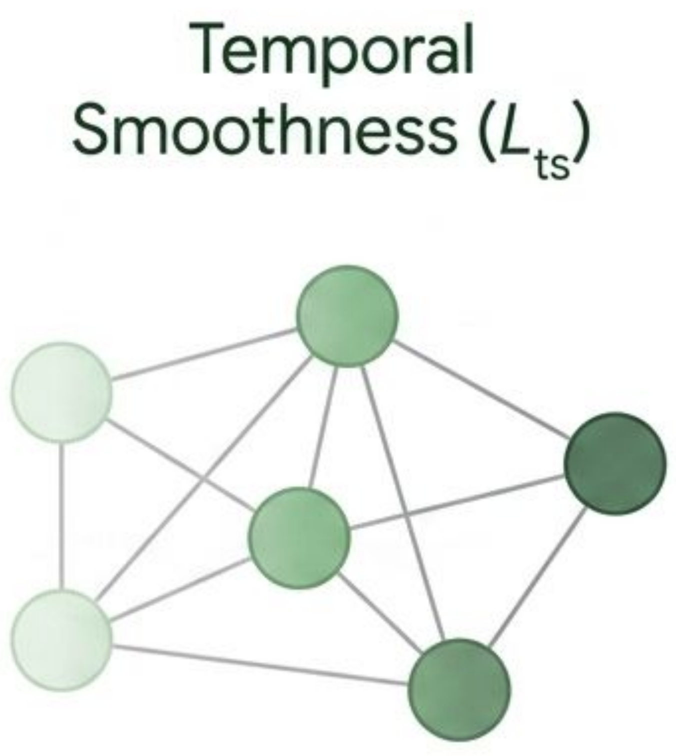


> The residual head learns t_i supervised by the standard VAE time via Mean Squared Error.

$$\mathcal{L}_{\text{time}} = \left\| t_{\text{head}} - \underbrace{\text{sg}\left[t_{\text{ODE}} \right]}_{\text{stop-gradient}} \right\|^2$$

Contribution 03

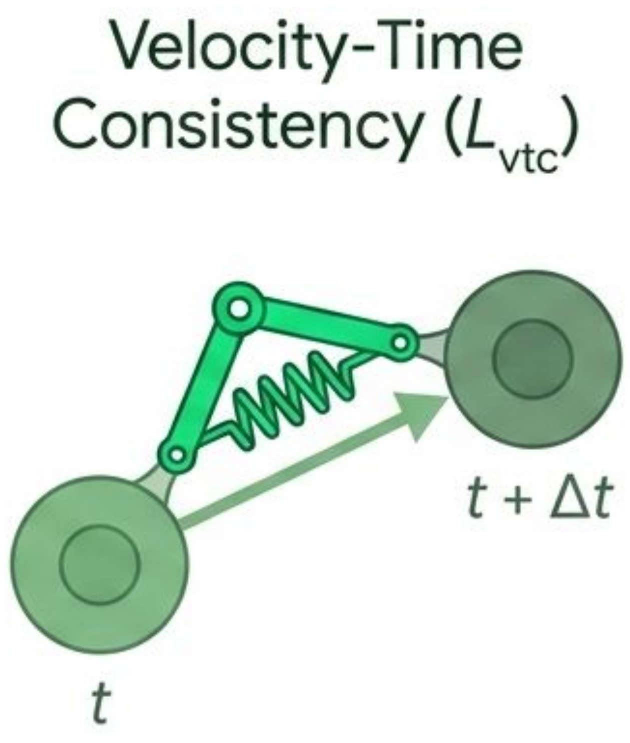
Two more geometric regularisers



kNN in latent space (k=10):

$$\mathcal{L} = \frac{1}{k} \sum_{j \in \mathcal{N}(i)} (t_i - t_j)^2$$

- Cells that look alike should sit at similar points in the trajectory.



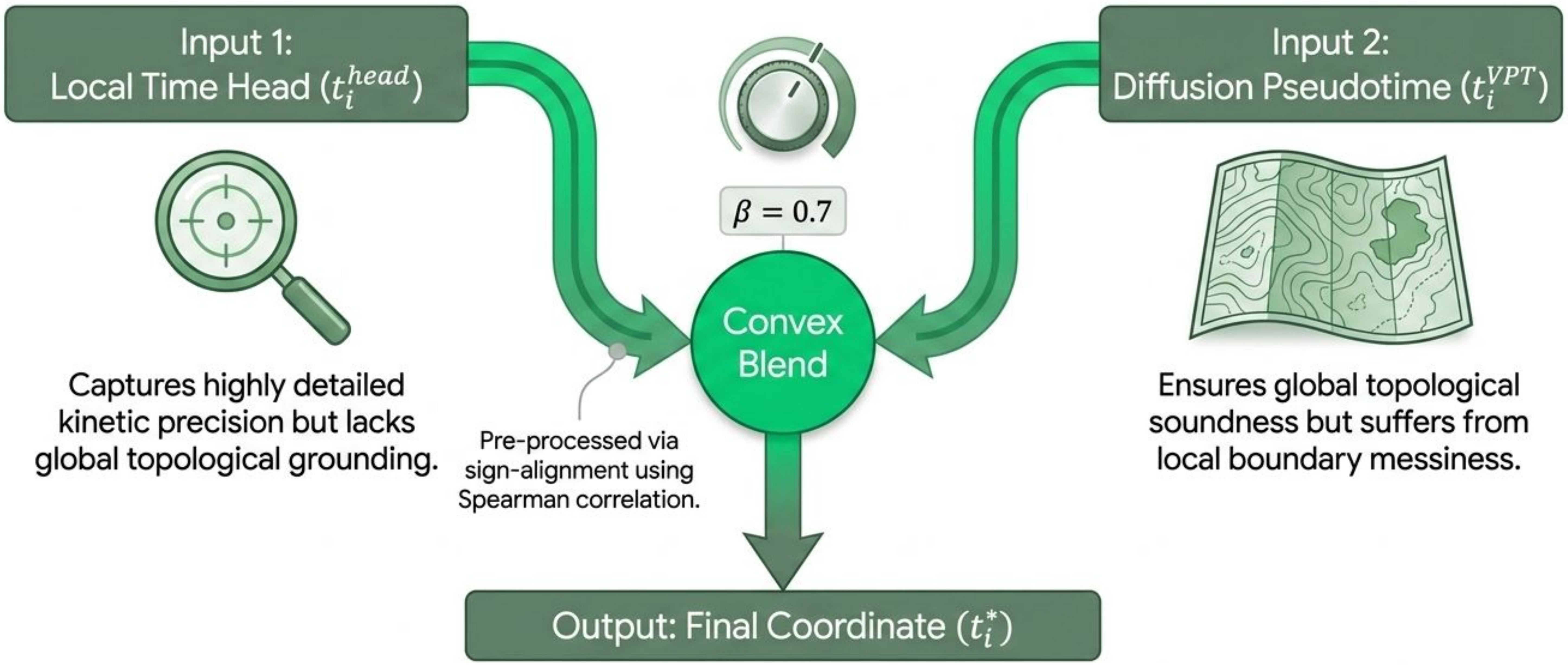
A Pairwise hinge penalty (ReLU-cos)

$$\mathcal{L} = \mathrm{relu}\big(-\cos(v_i,\ s_j - s_i)\big)$$

- Penalises velocity arrows that run backwards in time.

Contribution 04

Velocity–pseudotime blending



$$t_i^\star = \beta \cdot t_i^{\mathrm{VPT}} + (1-\beta) \cdot t_i^{\mathrm{head}}, \qquad \beta = 0.7$$

A rigorous benchmarking arena across 5 biological systems.

Biological Diversity
(Datasets)


Pancreatic Endocrinogenesis


Bone Marrow Haematopoiesis


Forebrain Neuronal Lineages


Chromaffin Sympathetic System


scEU Organoid

Evolution of the Field
(Baselines)

Velocyto
(Linear regression)

↓

scVelo dynamical
(Analytical ODE, EM)

↓

VeloVI
(Generative VAE)

↓

TFvelo
(Static TF linear regression)

Evaluation Dimensions
(Metrics)

Directional Accuracy

CBDir

ICCoH

Field Consistency

VelConf

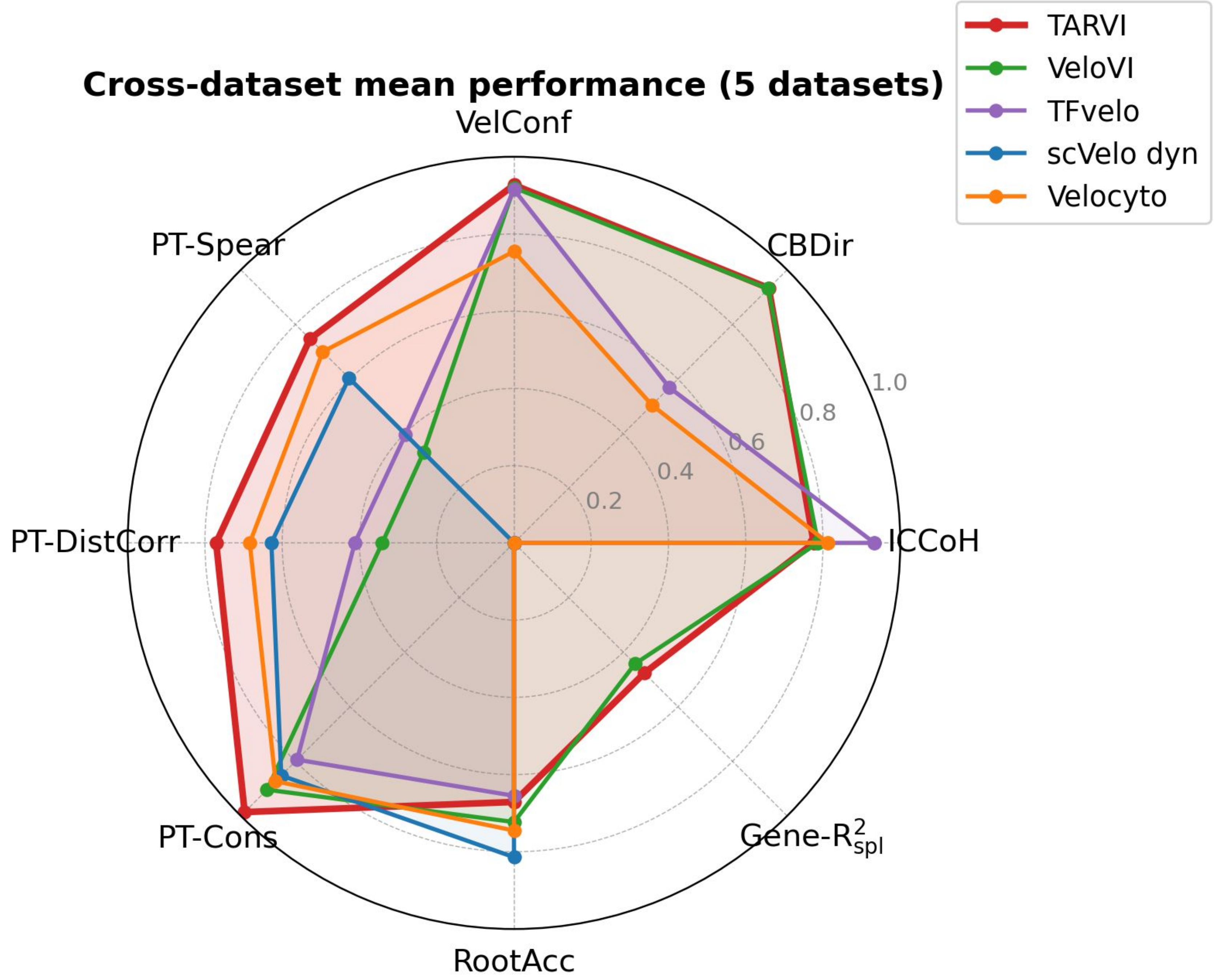
Tempora'Ordering

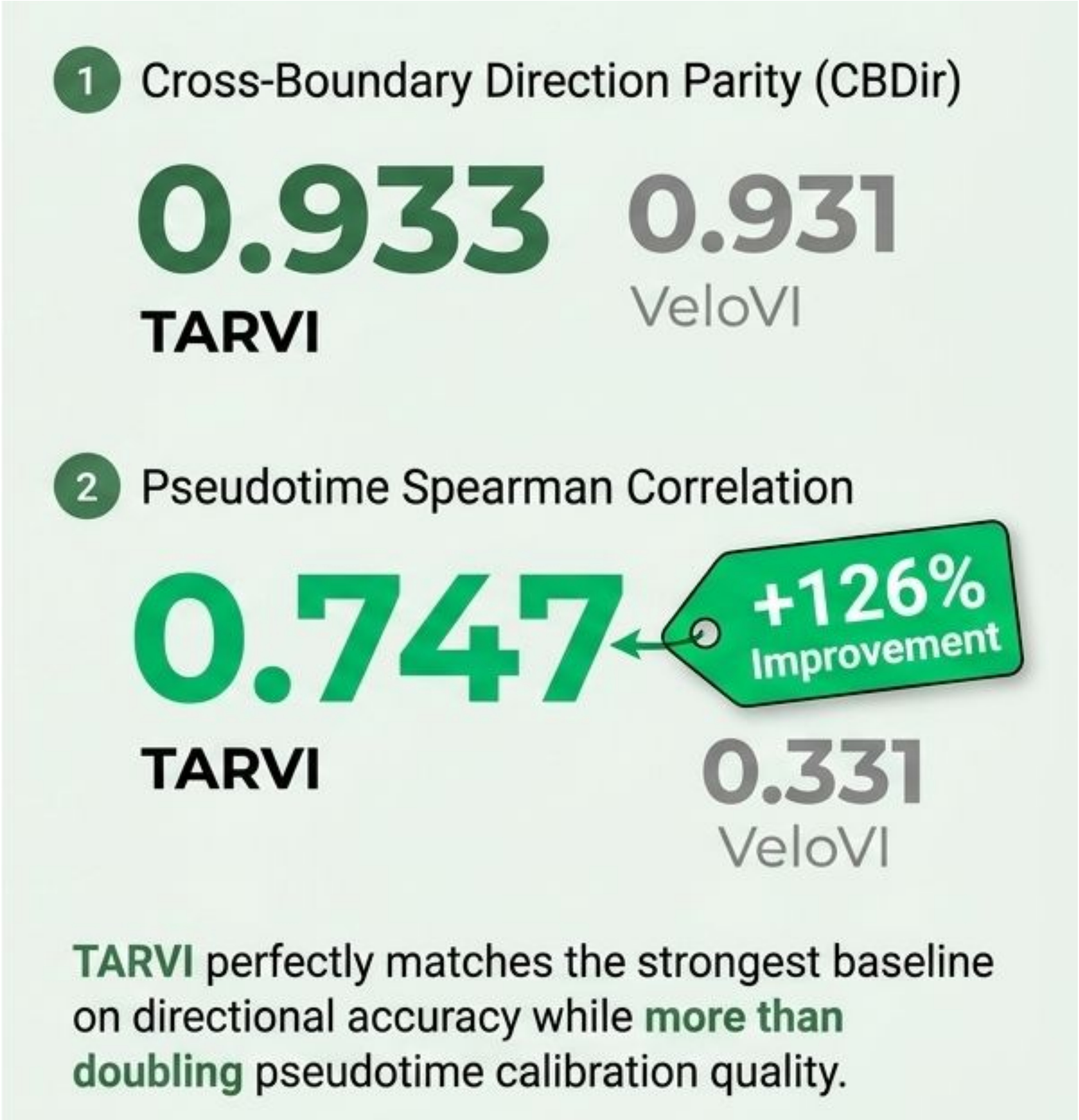
PT-Spear

PT-DistCorr

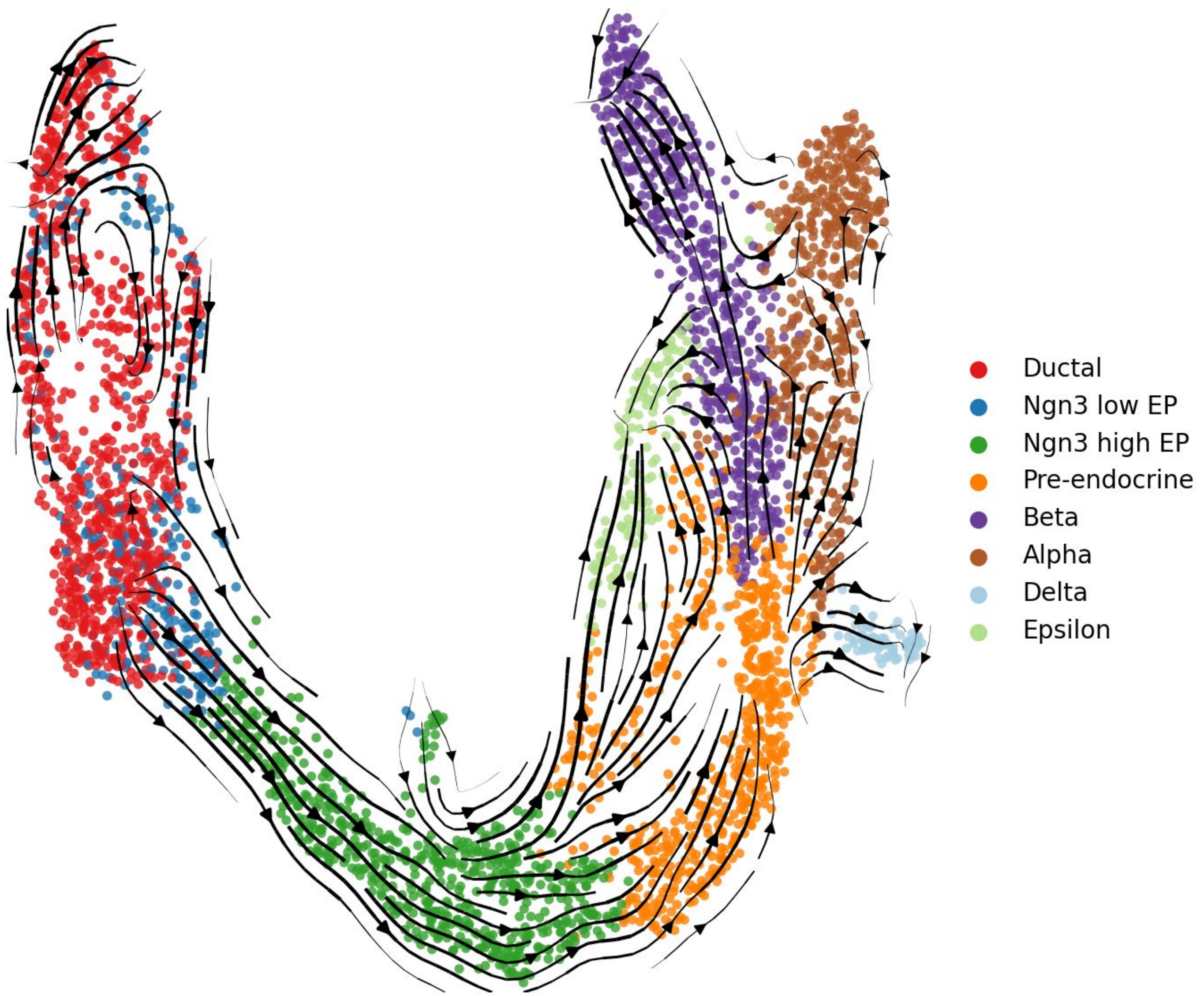
RootAcc

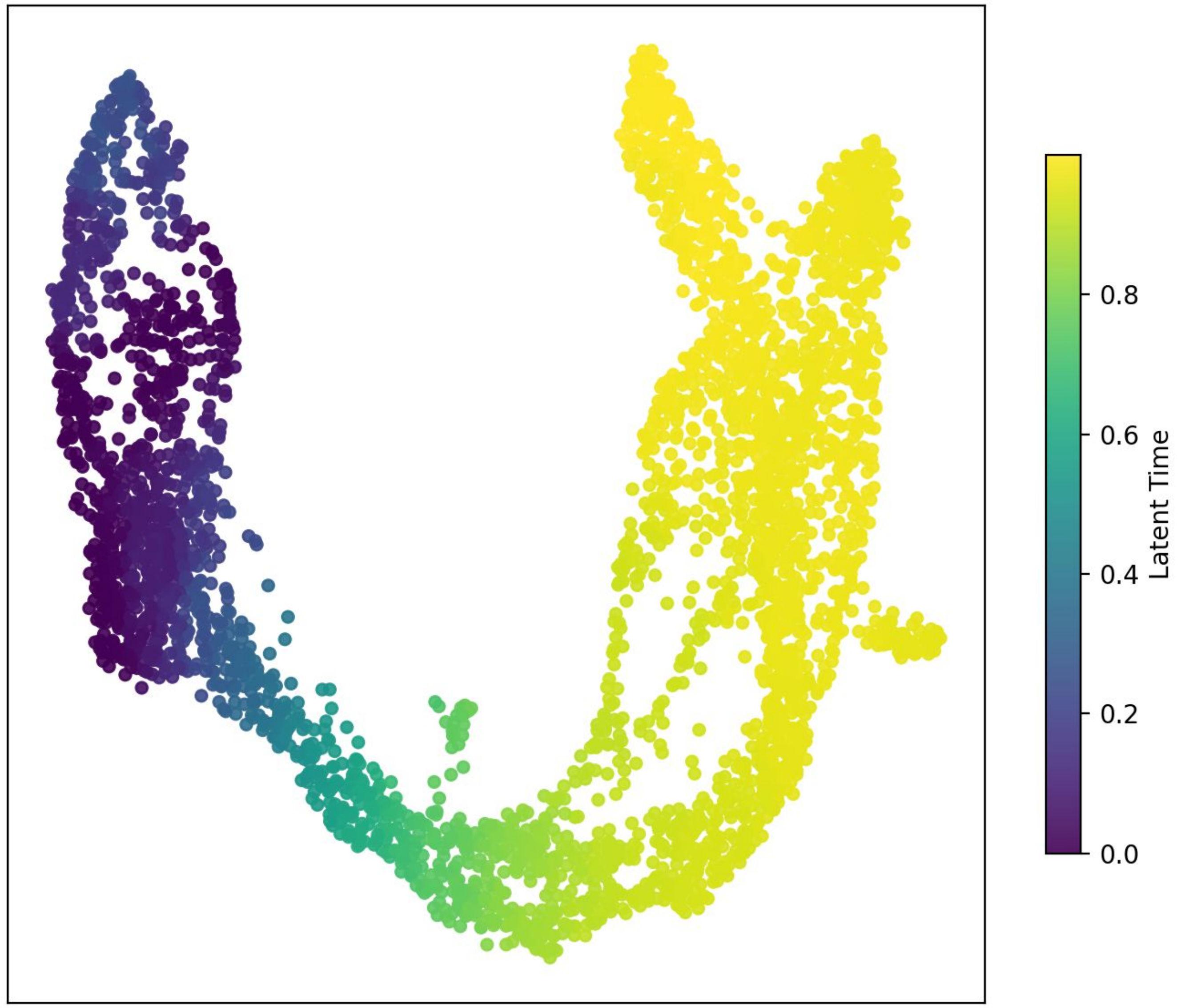
Quantitative Performance





Qualitative Performance





Continuous Directional Flow

Smooth Latent Time Gradient

Thank you

Questions?

Open Source
github.com/chandula00/TARVI